

ECON-UA 12

Intermediate Macroeconomics

Jaroslav Borovička, NYU, Fall 2013

Topic 4

The Solow growth model

Lecture objectives and reading

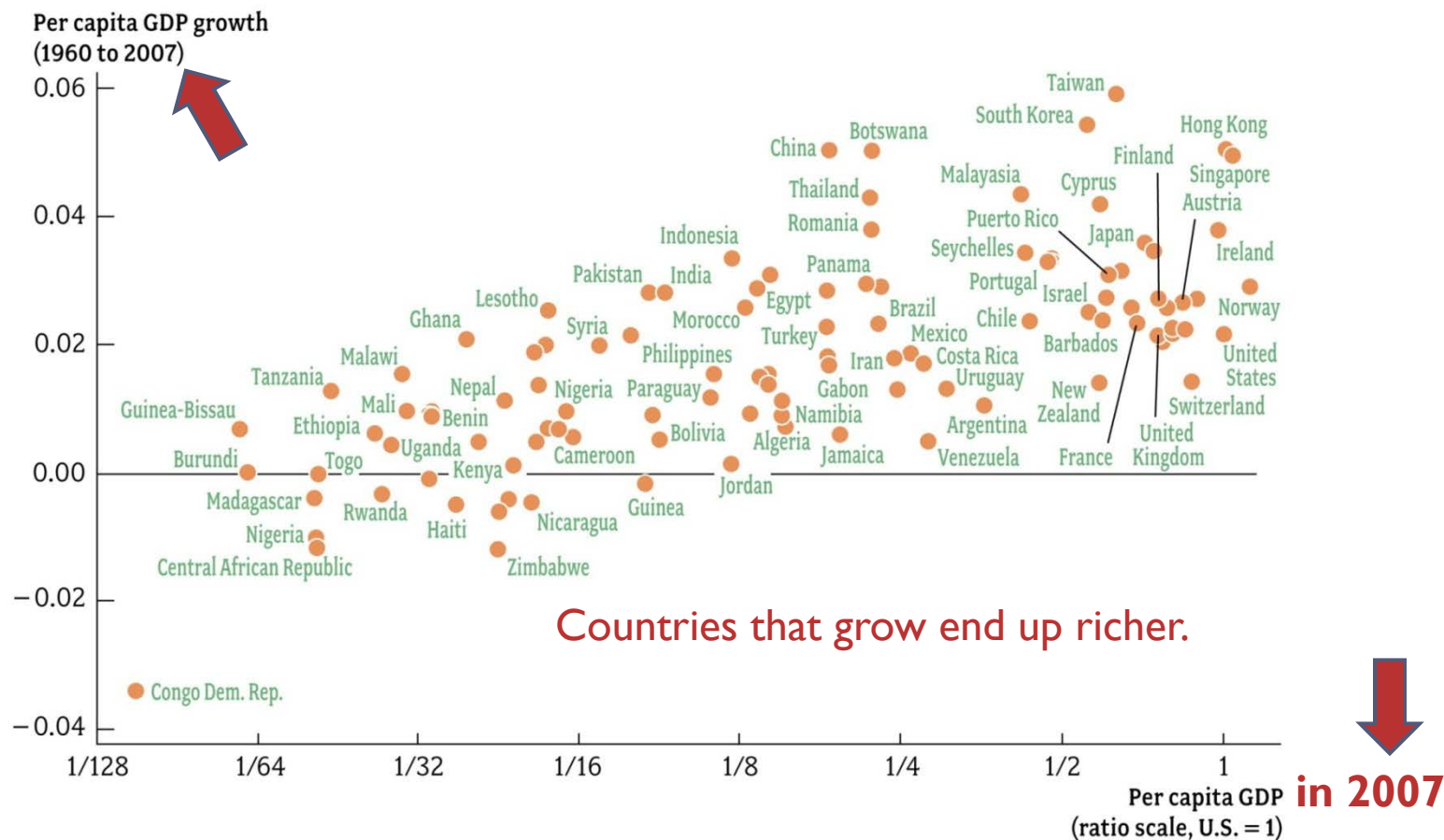
- ▶ Previous lectures – a model of production
 - ▶ Studied whether differences in capital per capita explain differences in output per capita.
 - ▶ But did not investigate why countries differ in the level of capital per capita.
- ▶ Solow growth model – a model of capital accumulation
 - ▶ Q: What generates cross-country differences in the level of capital per capita? Why do economies grow?
 - ▶ A: In the model, the answer is different saving rates.
 - ▶ What will remain unexplained is why should countries have different saving rates in the first place.
- ▶ Tools
 - ▶ Capital accumulation equation
 - ▶ Transition dynamics and steady state
- ▶ Reading
 - ▶ Jones – Chapter 5 + additional materials on course website

Outline

1. Empirical evidence
2. Setting up the Solow growth model
3. Saving, investment, and the real interest rate
4. Solving the Solow growth model
 1. Transition dynamics
 2. Steady state analysis, comparative statics
5. Predictions of the model and empirical evidence
6. Lessons and deficiencies

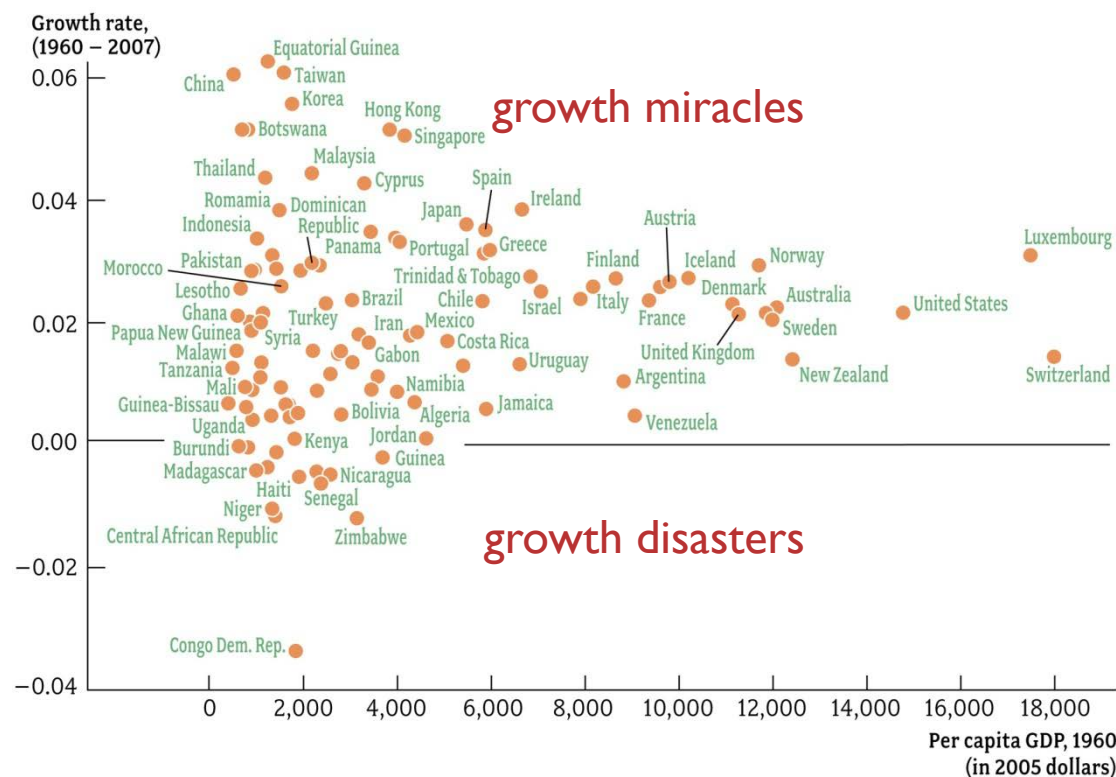
A cross-country comparison – ex post

- ▶ Countries have different growth rates and different levels of GDP per capita. But **why?**



A cross-country comparison – ex ante

- ▶ What about the role of initial conditions? Do countries that start poor catch up? Is there **convergence**?



1. Countries that started rich in 1960 grew at stable rates around 2%.
2. Growth rates of countries that started poor were all over the place.
3. **Growth miracles and growth disasters.**
4. **No convergence ?**

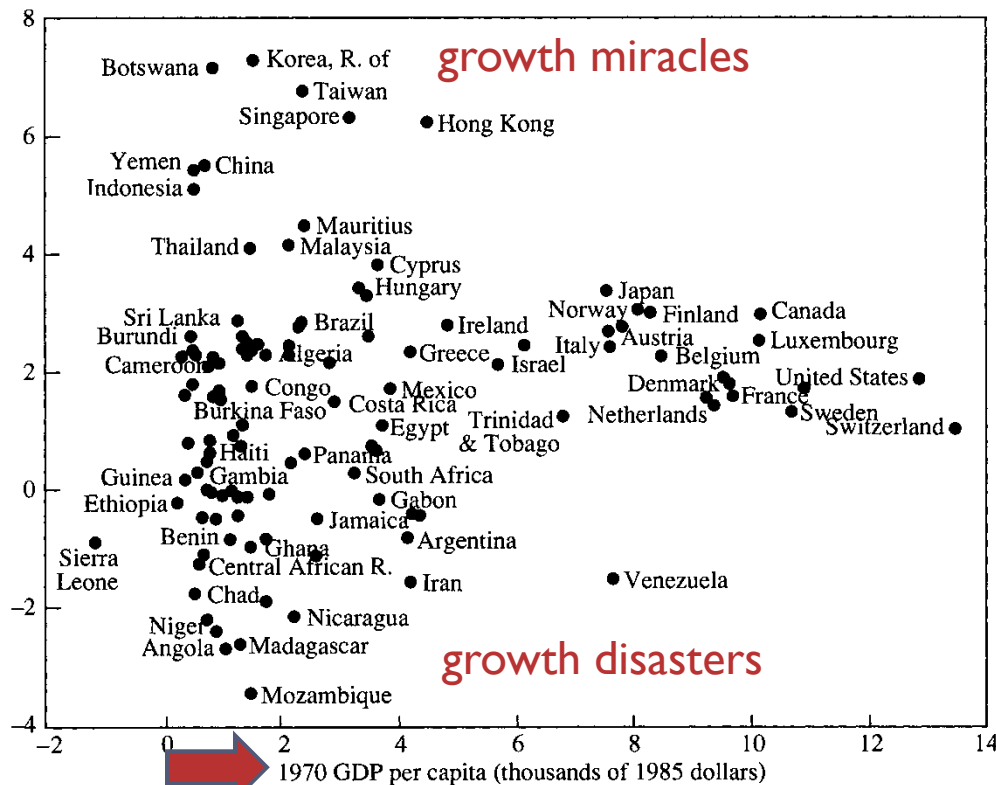


A cross-country comparison – ex ante

- ▶ What about the role of initial conditions? Do countries that start poor catch up? Is there **convergence**?

Figure 3. Growth and Initial Income, All Economies, 1970–89

Annual growth
per capita (percent)



Source: Version 5.5 of the data in Summers and Heston (1991) and World Bank (1994d).

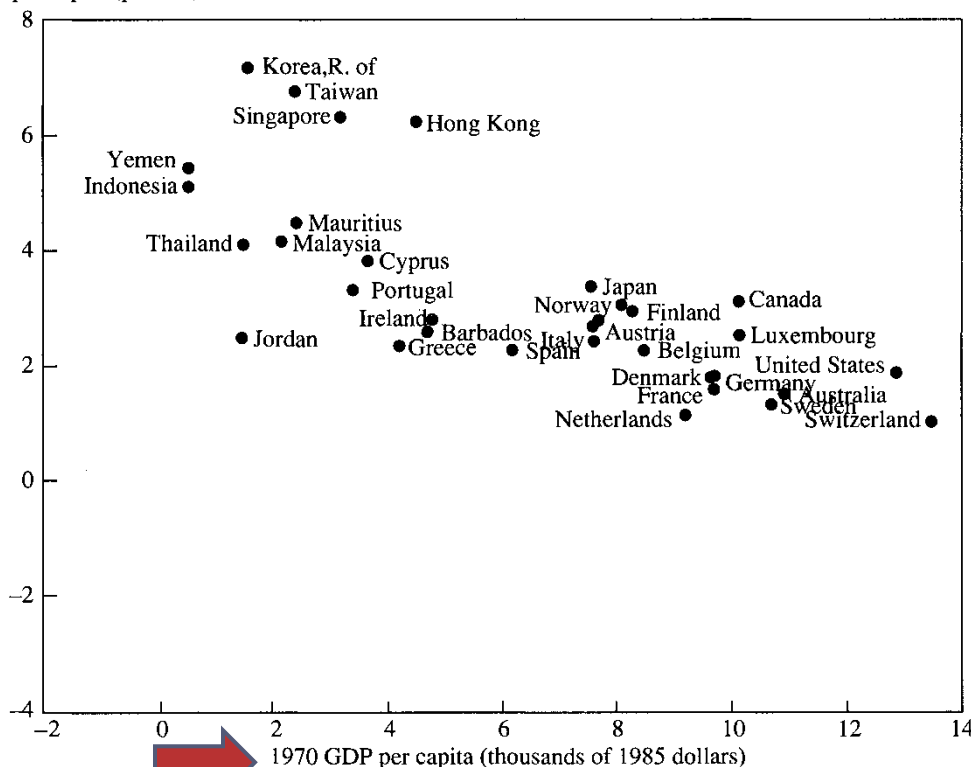
1. Countries that started rich in 1970 grew at stable rates around 2%.
2. Growth rates of countries that started poor were all over the place.
3. **Growth miracles and growth disasters.**
4. **No convergence ?**

A cross-country comparison – ex ante

- ▶ Let us look only at countries who opened their economies (probably also conducted other sensible policies).

Figure 4. Growth and Initial Income, Open Economies, 1970–89

Annual growth
per capita (percent)



Source: Authors' calculations using version 5.5 of the data in Summers and Heston (1991).

1. Poor countries grow much faster.

2. **Convergence!**

We will see such dynamics in our model. Countries grow fast initially, then growth slows down.

However, in our model, **growth will stop at some point**. This is a deficiency.

Example: South Korea and the Philippines

- ▶ **1960: very similar countries**
 - ▶ Per capita GDP around \$1500
 - ▶ Population: around 25 million, 1/2 working age
 - ▶ Similar sectoral composition (industry, agriculture)
 - ▶ College enrollment: South Korea 5%, Philippines 13%
- ▶ **Sharp divergence between 1960 and 2009**
 - ▶ Growth: South Korea = 5.4%, Philippines = 1.6% per annum
 - ▶ Per capita GDP in 2009
 - ▶ South Korea \$25,000
 - ▶ Philippines \$3,000
- ▶ **What happened?**

Why do countries grow?

- ▶ Solow growth model (Solow 1956 – Nobel Prize 1987)
 - ▶ This model continues to use the idea of **capital accumulation**.
 - ▶ **Chapter 4**: differences in capital per capita across countries as given.
 - ▶ We learned to which extent capital matters to explain differences in output ...
 - ▶ ... but did not learn about where the differences in capital come from.
 - ▶ **This chapter**: explain them through saving and investment that leads to capital accumulation.
- ▶ Moving from a static to a **dynamic model**
 - ▶ Country starts with a given amount of capital = **initial condition**.
 - ▶ It accumulates it over time = moves along **a transition path**...
 - ▶ ... and converges to a constant level of capital in the long run = **steady state**.
 - ▶ This steady state will depend on the **saving rate** = fraction of income not consumed but saved and used as investment

1.

2.

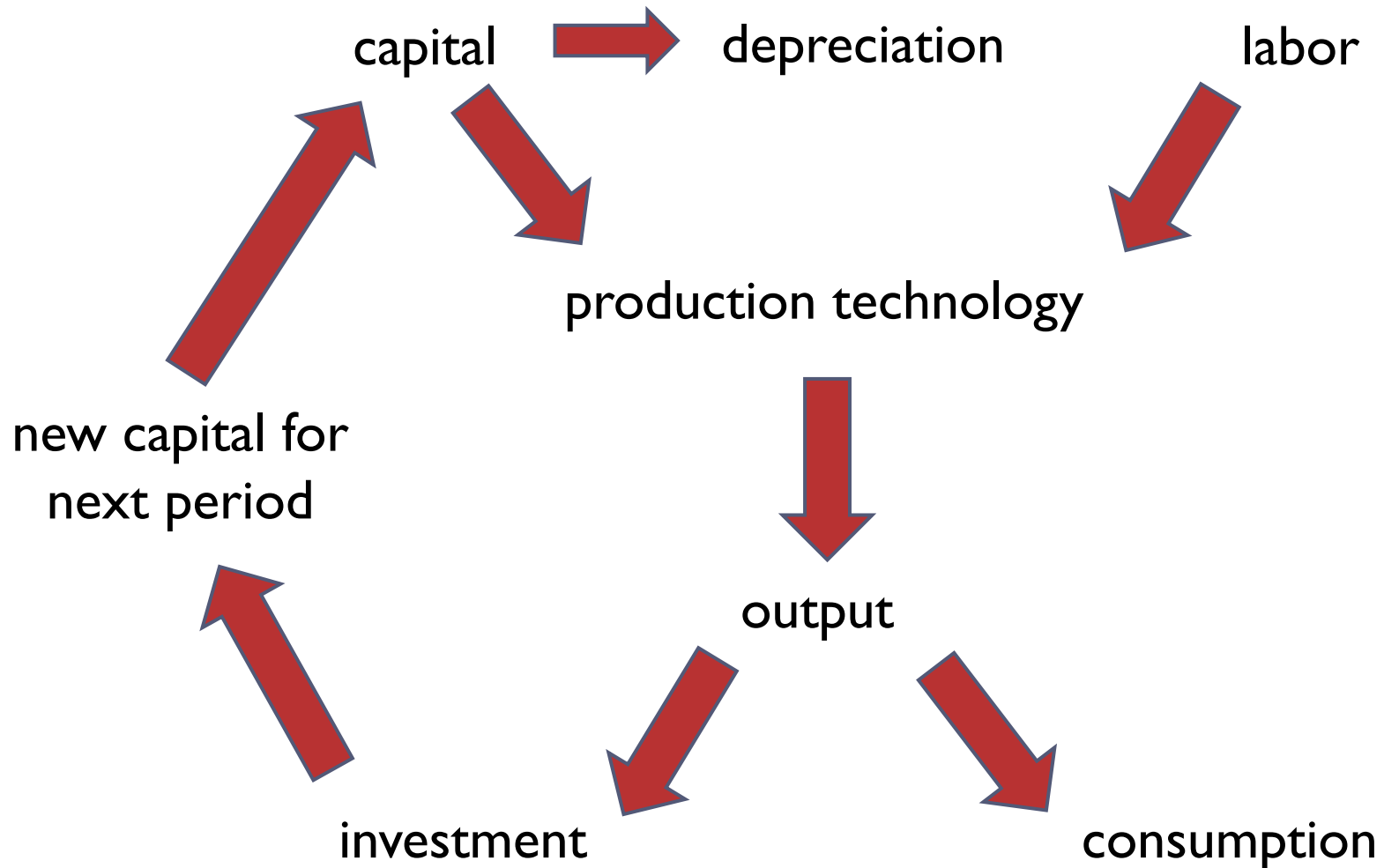
4.

3.

Outline

1. Empirical evidence
2. Setting up the Solow growth model
3. Saving, investment, and the real interest rate
4. Solving the Solow growth model
 1. Transition dynamics
 2. Steady state analysis, comparative statics
5. Predictions of the model and empirical evidence
6. Lessons and deficiencies

Framework of the model – sketch



Stock and flow variables

There are two types of variables in the model.

- ▶ **Stock variables**

- ▶ A quantity that is measured at an instant in time.
- ▶ Stock of capital (buildings, ...), population, ...

- ▶ **Flow variables**

- ▶ A quantity that is measured over a period of time.
- ▶ Output – produced over the course of a year.
- ▶ Investment, depreciation, ...

- ▶ **Flow of investment adds to the stock of capital.**

Framework of the model

- ▶ Discrete time: $t = 0, 1, 2, 3, \dots$ (indexed, e.g., in years)
 - ▶ Infinite horizon = time never ends, t continues to infinity
 - ▶ Economy starts at $t = 0$ with capital K_0
- ▶ In every period t , output is produced using Cobb-Douglas technology: $Y_t = F(K_t, L_t) = AK_t^\alpha L_t^{1-\alpha}$ ($\alpha = \frac{1}{3}$)
 - ▶ Assume labor force is fixed: $L_t = \bar{L}$
- ▶ Resource constraint: $Y_t = C_t + I_t$
 - ▶ Output divided between consumption and investment
- ▶ Capital accumulation equation: $K_{t+1} = K_t + I_t - \delta K_t$
 - ▶ Investment adds to capital, capital depreciates at rate δ
- ▶ Savings function
 - ▶ Households save a constant fraction of income: $I_t = sY_t$

Framework of the model

► Variables

► Capital K_t

► Labor L_t

► Output Y_t

► Consumption C_t

► Investment I_t

5 equations, 5 variables



► Parameters

► Depreciation rate δ (0.1)

► Savings rate s (0.2-0.5)

► Size of labor force L

► Total factor productivity A

► Initial amount of capital K_0

► Equations

► Technology:

$$Y_t = AK_t^\alpha L_t^{1-\alpha}$$

► Resource constraint:

$$Y_t = C_t + I_t$$

► Capital accumulation:

$$K_{t+1} = K_t + I_t - \delta K_t$$

► Saving/investment function:

$$I_t = sY_t$$

► Fixed labor force:

$$L_t = \bar{L}$$

Outline

1. Empirical evidence
2. Setting up the Solow growth model
3. Saving, investment, and the real interest rate
4. Solving the Solow growth model
 1. Transition dynamics
 2. Steady state analysis, comparative statics
5. Predictions of the model and empirical evidence
6. Lessons and deficiencies

How to think about saving and investment

▶ Example: a farm

- ▶ A farm starts in winter with seed grains (capital). Some of the grains get spoiled during the winter (depreciation).
- ▶ In spring, the crop is planted (used in production).
- ▶ In fall, harvest (output) is collected.
 - ▶ Farmer puts aside the original amount of grains (capital).
 - ▶ A fraction of the remaining harvest is used for consumption.
 - ▶ The rest is stored and increases the amount of capital (investment).

▶ Observe that there is no money in this economy.

- ▶ Households do not save using money. They save by accumulating real resources (capital).
- ▶ Here, they can decide whether to use grains for consumption or as investment (**perfectly substitutable goods**).
- ▶ In reality, agents would have to decide beforehand whether to produce **investment goods** or **consumption goods**.

Saving and investment decision

- ▶ How do households decide how much to save?
 - ▶ Here, we assume that households save a constant fraction of income.
 - ▶ In a more complicated model, households would compare the benefits of consuming the good today with the benefits of saving and consuming later (example: farmer and grains).
 - ▶ Consumption smoothing
- ▶ Observe that an economy can grow faster if it increases its saving rate.
 - ▶ Households have to sacrifice current consumption = tradeoff.
 - ▶ Example: Chinese saving rate is incredibly high. Future generations will be better off but current generations would benefit from more consumption today.

Role of the real interest rate

- ▶ How much should the household save?
 - ▶ That depends on the interest rate it receives on the savings.
 - ▶ Here, the household saves real resources = **real interest rate**.
 - ▶ See Chapters 15 and 16 for details.
- ▶ What is the real interest rate in the Solow model?
 - ▶ **Return on investment** = the return capital owners earn for saving a unit of output and renting it out to firms
 - ▶ **Rental price of capital** = price firms pay to capital owners
 - ▶ **Marginal product of capital** = amount an extra unit of capital earns in a firm (paid to capital owners)
 - ▶ **User cost of capital** = cost of maintaining one unit of capital, including opportunity cost
- ▶ The above terms all represent the same economic concept in this model.

Real interest rate

- ▶ Real interest rate is the same as in the model of production from Chapter 4.
 - ▶ Marginal product of capital equals the rental rate for capital (see lecture)

$$r = MPK = u = R + \delta - \frac{\Delta p_k}{p_k}$$

- ▶ For Cobb-Douglas

$$MPK = \alpha A \left(\frac{K}{L} \right)^{\alpha-1}$$

- ▶ As capital per capita grows, real interest rate decreases.
- ▶ It is less and less profitable to invest.

Outline

1. Empirical evidence
2. Setting up the Solow growth model
3. Saving, investment, and the real interest rate
4. Solving the Solow growth model
 1. Transition dynamics
 2. Steady state analysis, comparative statics
5. Predictions of the model and empirical evidence
6. Lessons and deficiencies

Solving the Solow growth model

- ▶ Combine equations to get

$$\Delta K_{t+1} \equiv K_{t+1} - K_t = sY_t - \delta K_t$$
$$Y_t = AK_t^{1/3} L^{2/3}$$

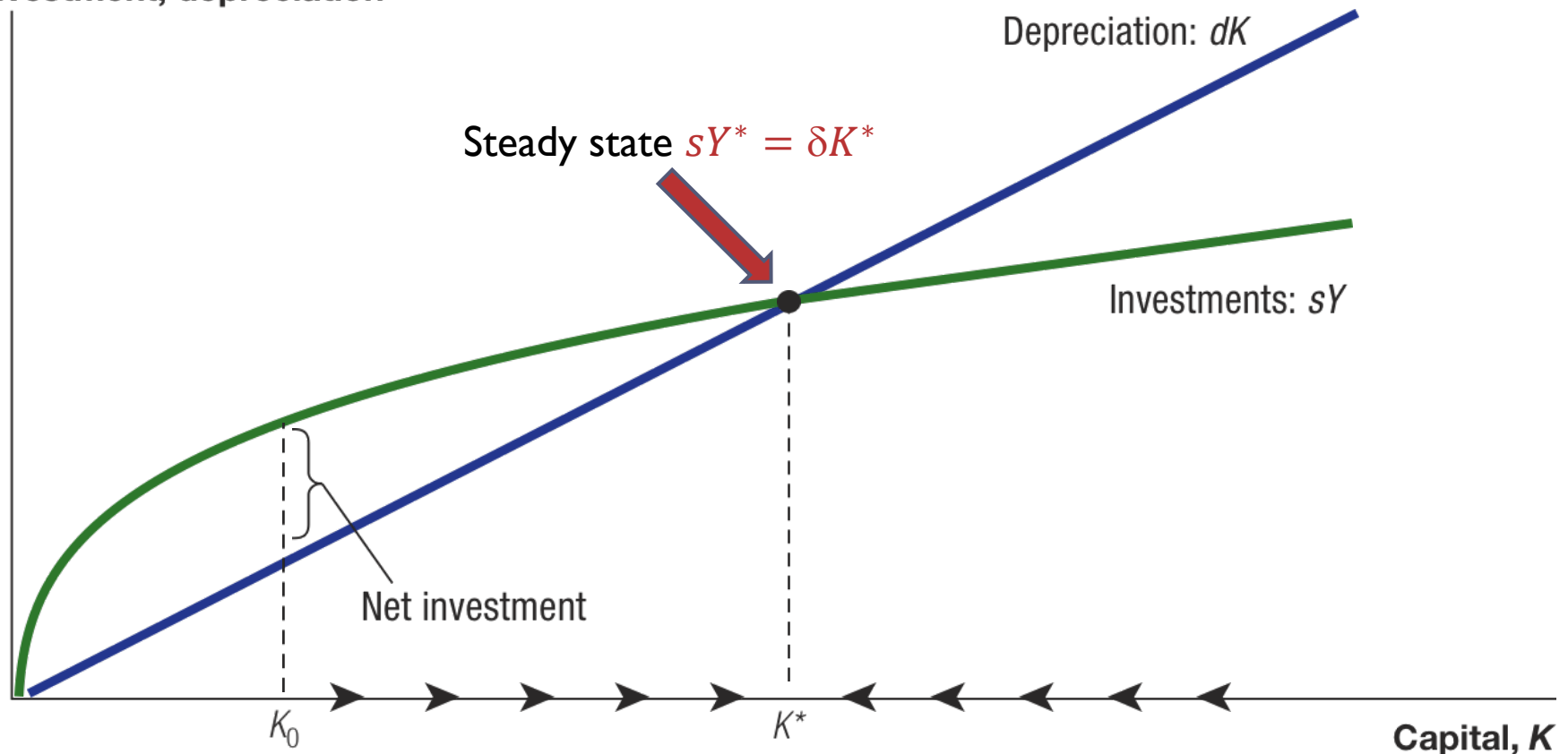
- ▶ Start with capital K_t
- ▶ Compute output Y_t
- ▶ Compute net investment $sY_t - \delta K_t$, which is the change in capital for next period ΔK_{t+1} . This yields K_{t+1} .
- ▶ We can plot the transition dynamics of K_t : **Solow diagram**
- ▶ Capital accumulation is at the heart of the model
 - ▶ The model explains how capital is accumulated.
 - ▶ Capital is an **endogenous variable** (explained within the model, as opposed to **exogenous**, imposed from outside).

Solow diagram

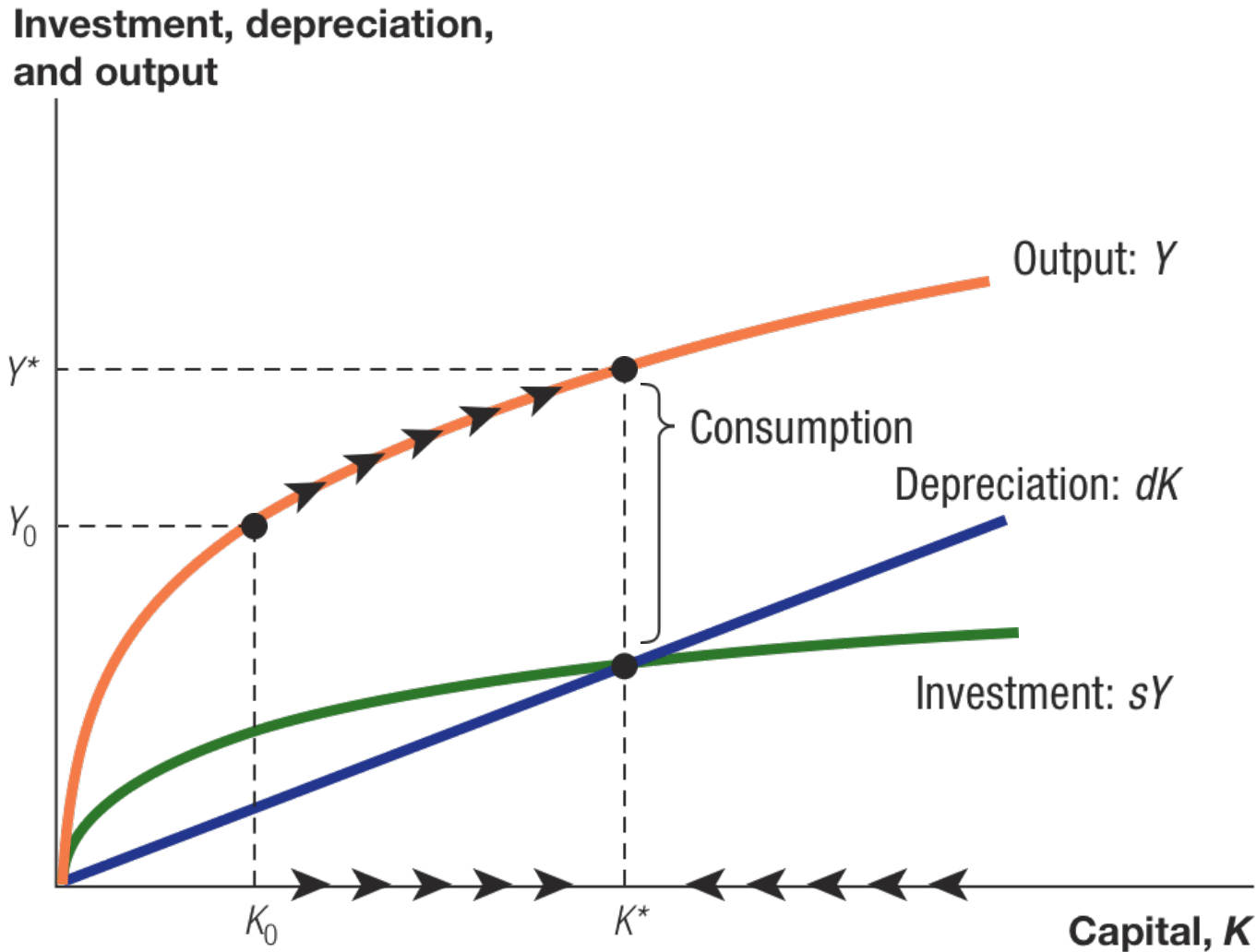
$$\Delta K_{t+1} = sY_t - \delta K_t$$

$$Y_t = AK_t^{1/3}L^{2/3}$$

Investment, depreciation



Solow diagram – adding consumption



Steady states vs. long-run growth

- ▶ Economy converges to a steady state
 - ▶ Capital accumulation alone **cannot generate long-run growth!**
 - ▶ Is the steady state every reached?
 - ▶ Only asymptotically.
- ▶ Why does growth vanish in the long run?
 - ▶ Depreciation is a constant share of capital.
 - ▶ But production exhibits **diminishing marginal product of capital** (diminishing returns to capital).
 - ▶ As capital increases, it is ever harder to replenish depreciated capital (think about the farm example, and about the computation of r).
 - ▶ **This is a disappointing result**, caused by diminishing returns to capital.
- ▶ Solution? Something else has to grow as well.
 - ▶ Population? Output per capita would still converge to a steady state.
 - ▶ Some factor included in TFP? Human capital, inventions, ... Chapter 6

Comparative statics

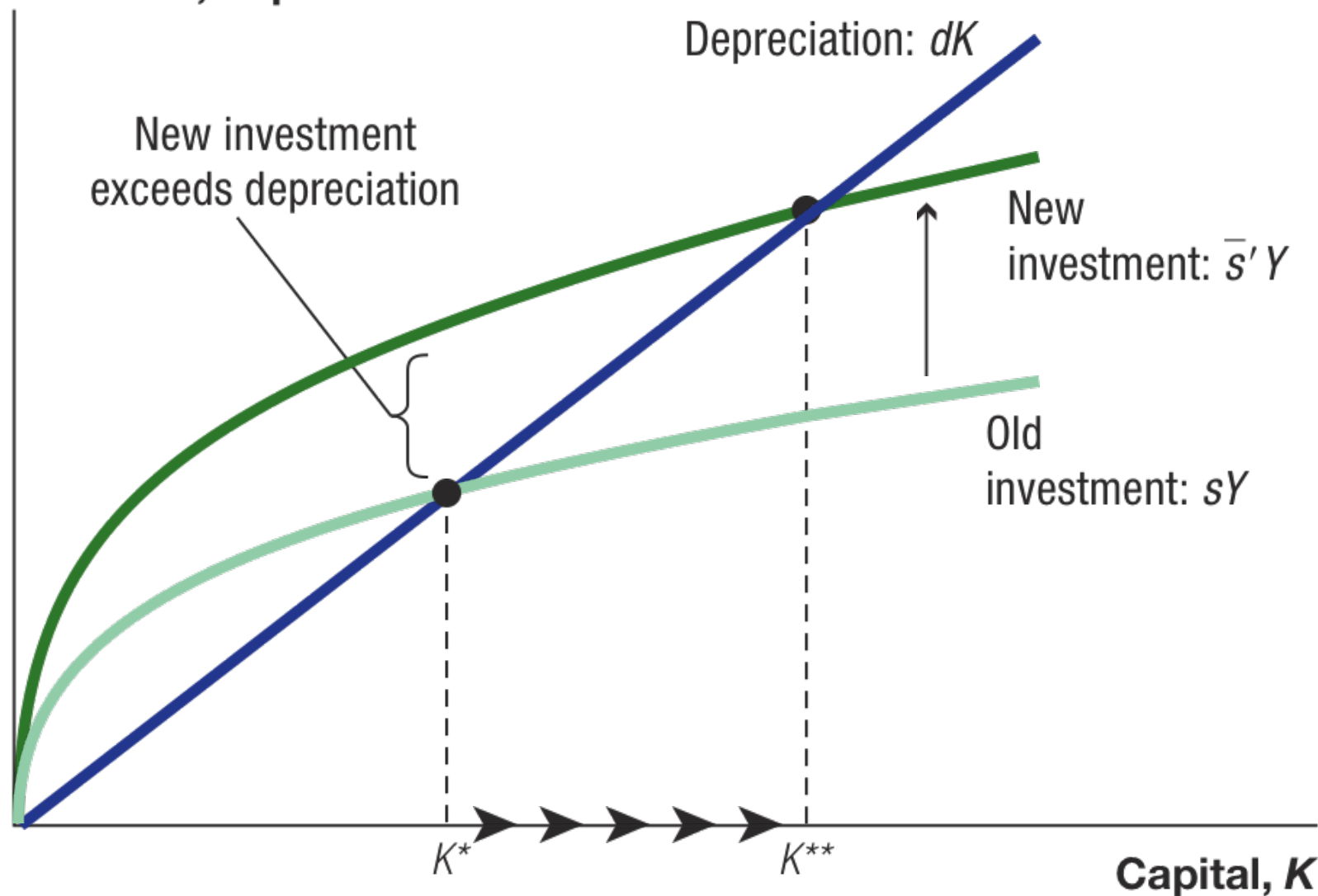
- ▶ A **comparative statics** exercise = change the value of a parameter, and observe how the solution changes
- ▶ Steady state solution: $sY^* = \delta K^*$
 - ▶ Solve for capital per capita: $\frac{K^*}{L} = \left(\frac{sA}{\delta}\right)^{\frac{1}{1-\alpha}}$
 - ▶ Solve also for output per capita!
- ▶ The steady state level of capital is
 - ▶ Increasing in saving rate
 - ▶ Increasing in total factor productivity
 - ▶ Decreasing in the rate of depreciation
- ▶ What is the economic interpretation of these results?
- ▶ Is this the only steady state?

Transition dynamics

- ▶ Despite the disappointing convergence to a steady state, studying the model can still provide us with insights.
- ▶ Questions we can ask:
 - ▶ Imagine that the economy starts in the steady state, and a natural disaster destroys 20% of the capital stock. What happens then?
 - ▶ See case study in textbook.
 - ▶ Imagine that the economy starts in the steady state with a saving rate of 10%. What happens when the saving rate increases to 15%?
 - ▶ What happens when TFP in the economy suddenly increases by 10%? How does the answer differ from the result in the static model of production from Chapter 4?
 - ▶ Capital accumulation matters.

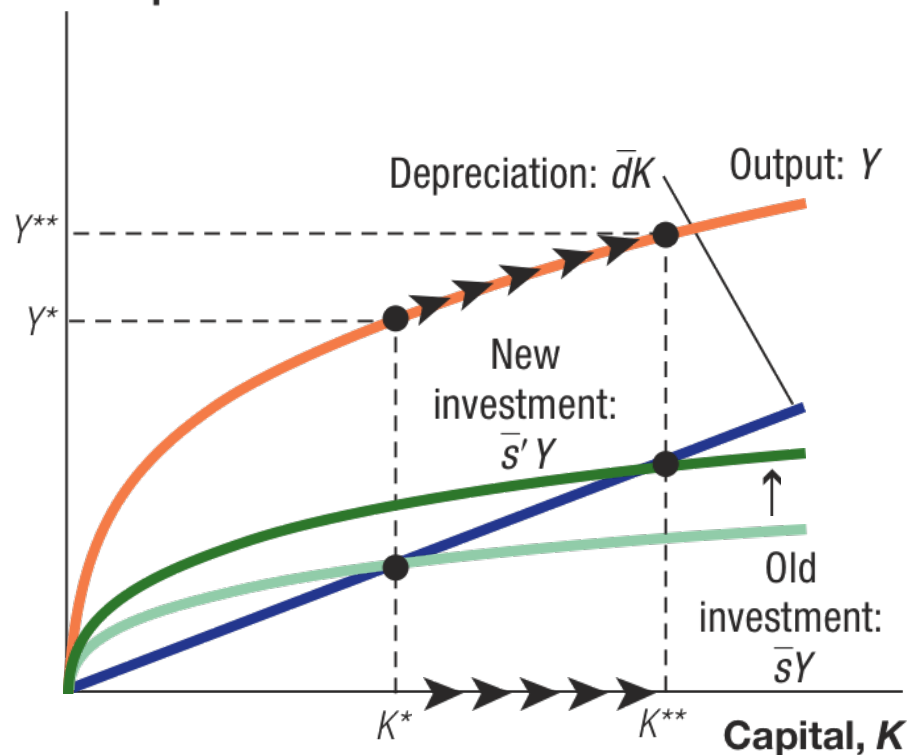
Transition dynamics – increase in saving rate

Investment, depreciation



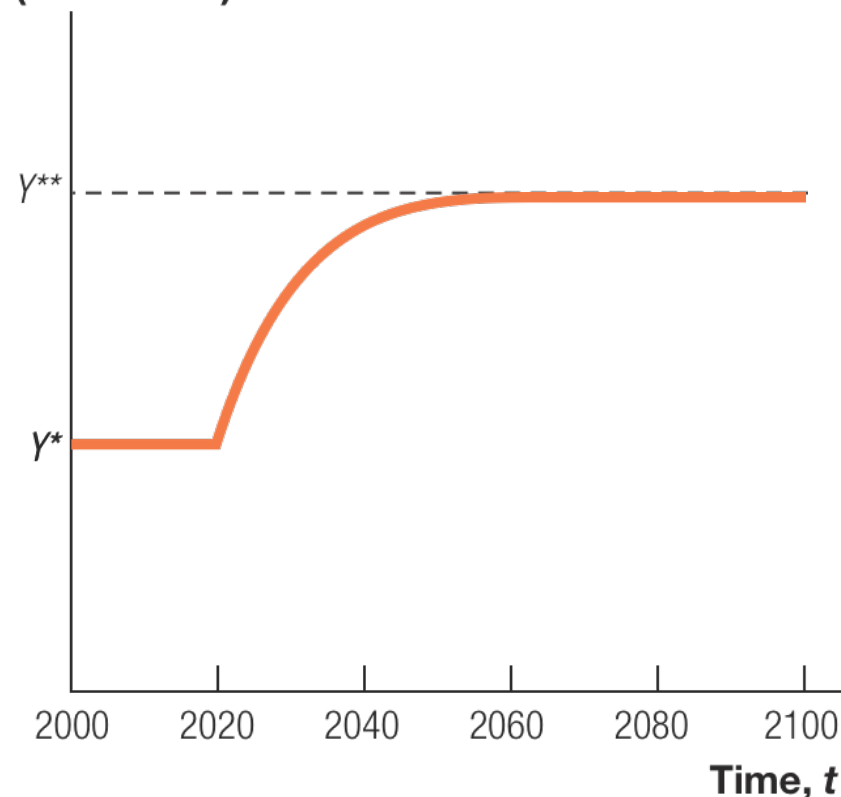
Transition dynamics – increase in saving rate

Investment, depreciation, and output



(a) The Solow diagram with output.

Output, Y
(ratio scale)



(b) Output over time.

Outline

1. Empirical evidence
2. Setting up the Solow growth model
3. Saving, investment, and the real interest rate
4. Solving the Solow growth model
 1. Transition dynamics
 2. Steady state analysis, comparative statics
5. Predictions of the model and empirical evidence
6. Lessons and deficiencies

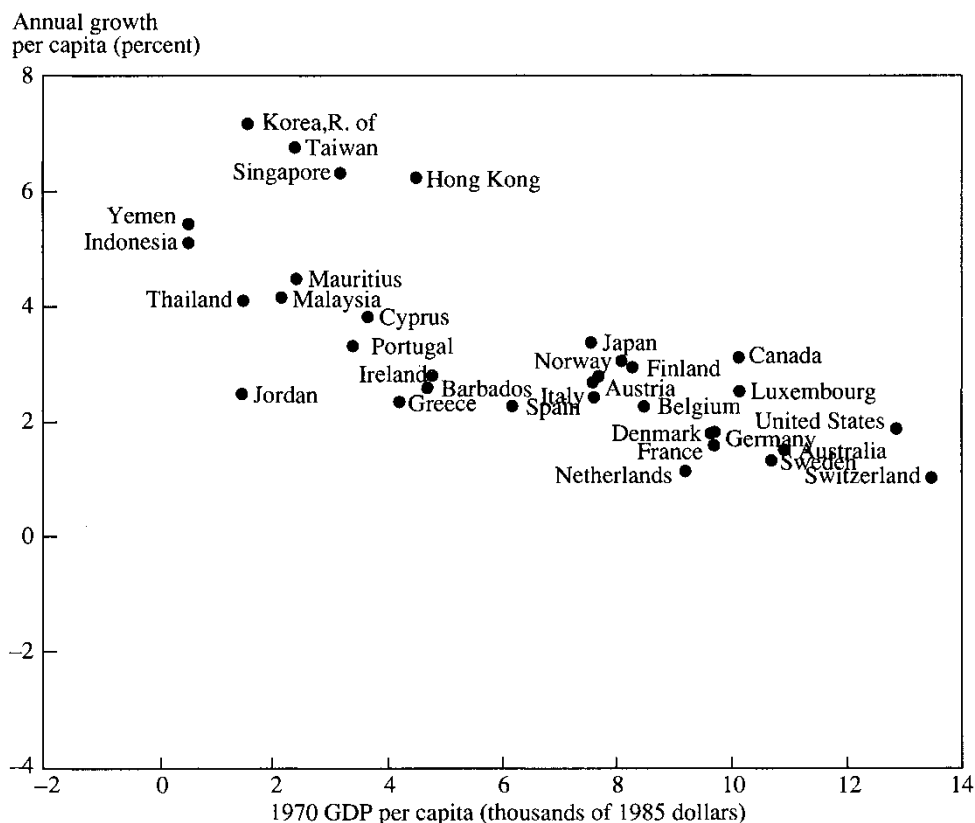
Predictions from the model

- ▶ **Distance from steady state**
 - ▶ The economy will grow the faster the farther below the steady state it is.
 - ▶ Prediction: Catching up.
- ▶ **Growth generated by capital accumulation**
 - ▶ Growth episodes accompanied by high rates of investment.
- ▶ **High permanent saving rates should lead to accumulation of large amounts of capital.**

Empirical evidence – open economies

► Recall graph from the beginning of the lecture

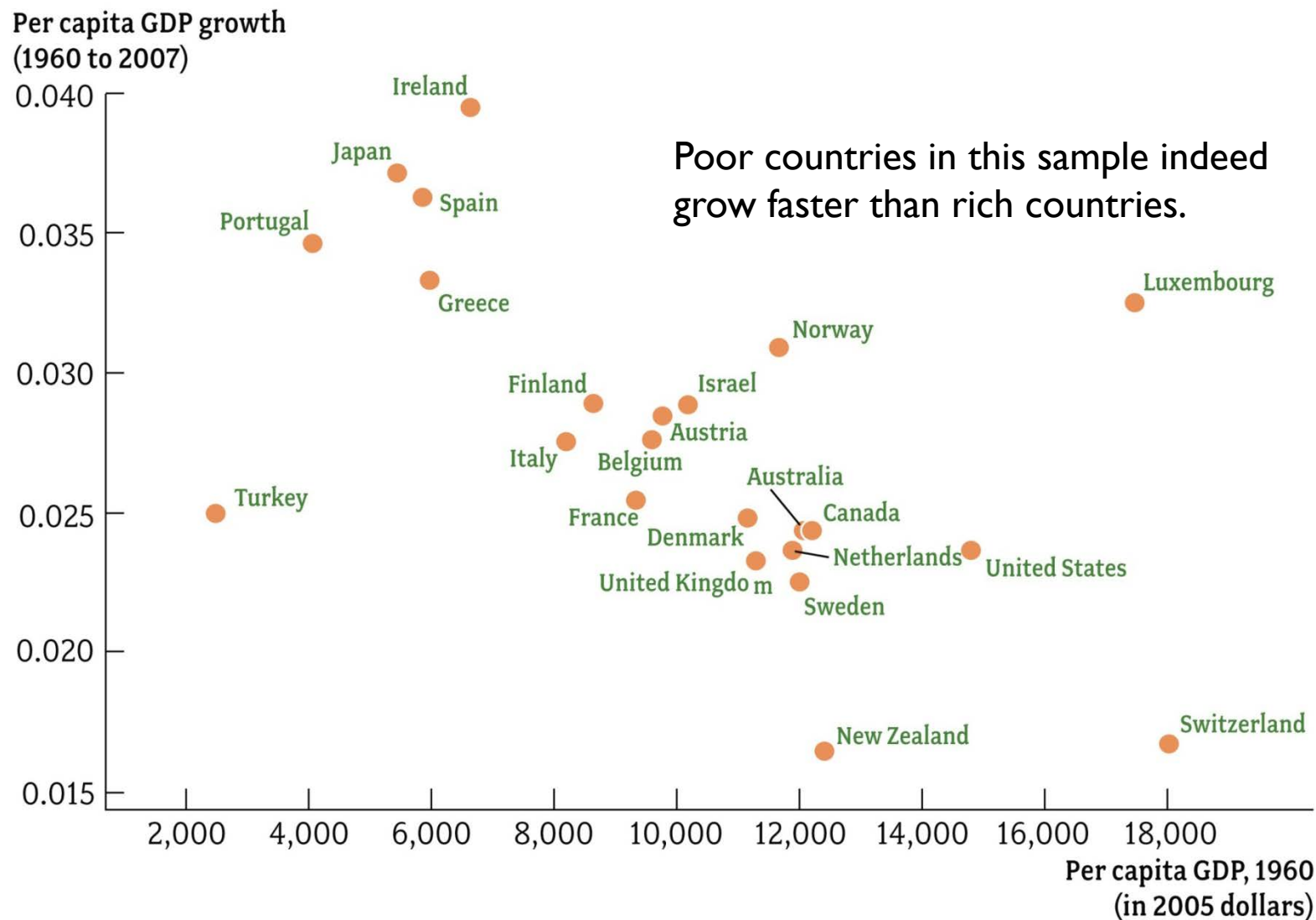
Figure 4. Growth and Initial Income, Open Economies, 1970–89



Source: Authors' calculations using version 5.5 of the data in Summers and Heston (1991).

Once countries start following plausible policies, the prediction of the Solow growth model seems to hold.

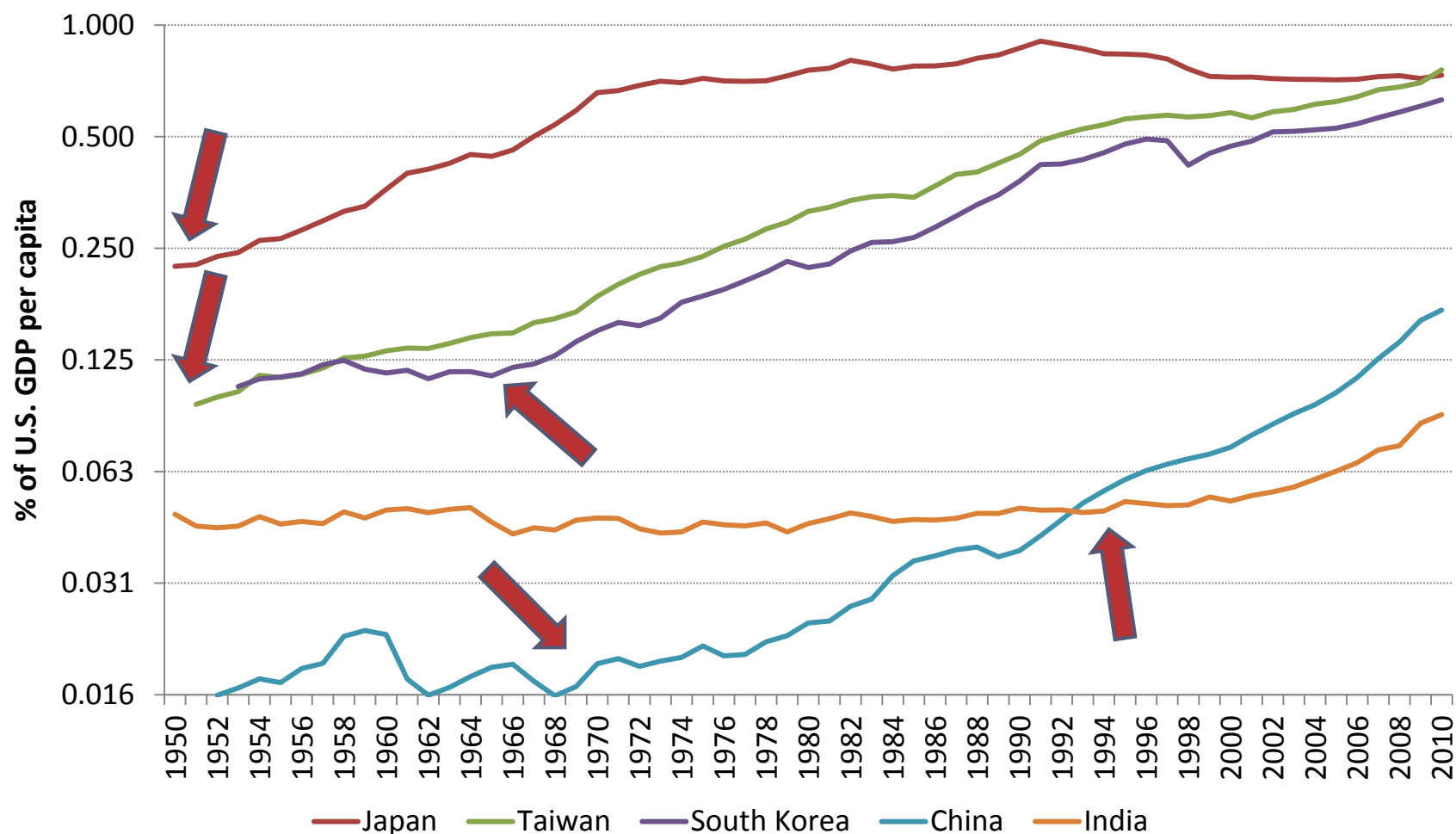
Empirical evidence – OECD countries



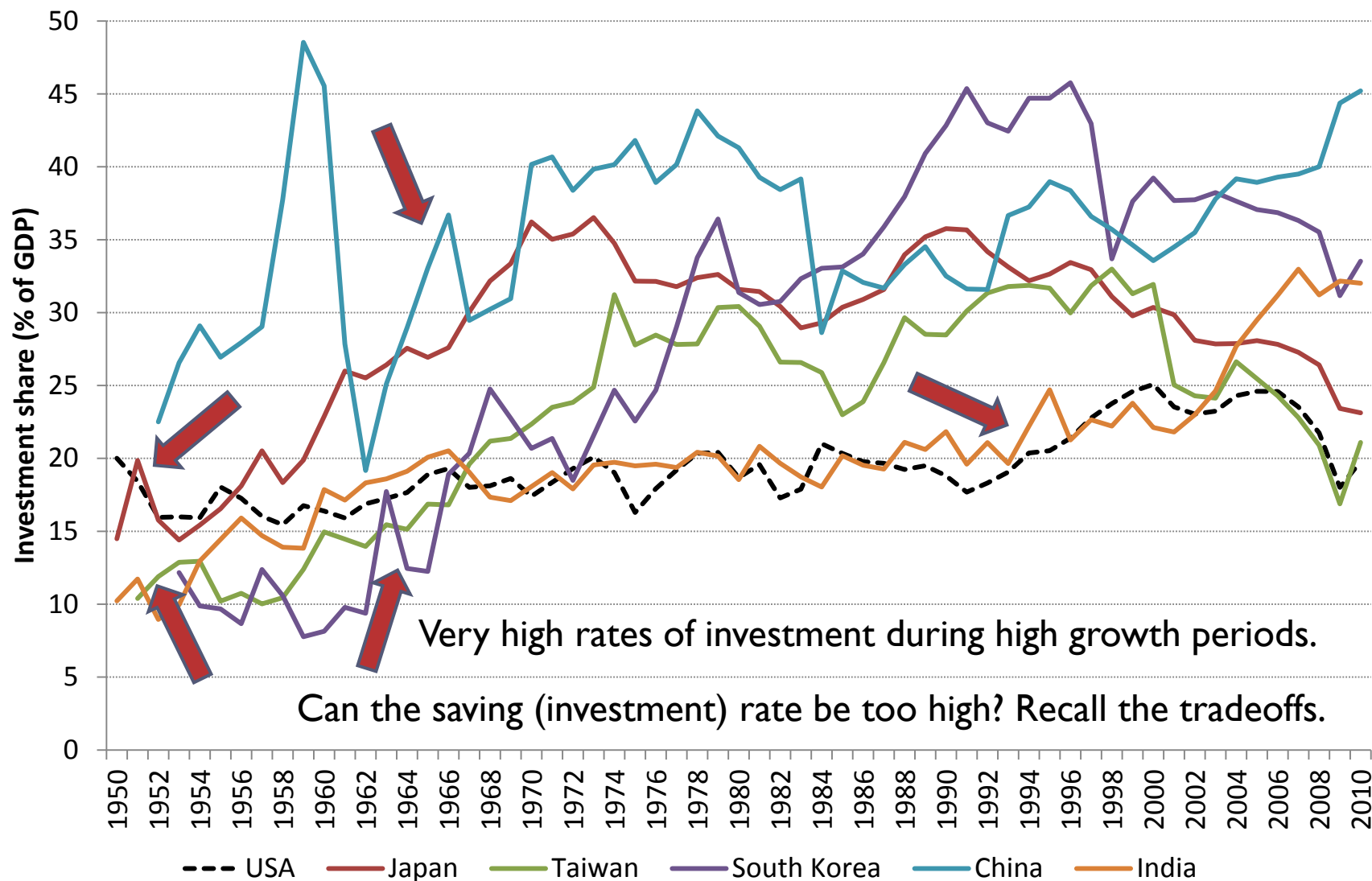


Empirical evidence – growth periods

► GDP per capita as a share of U.S. GDP per capita

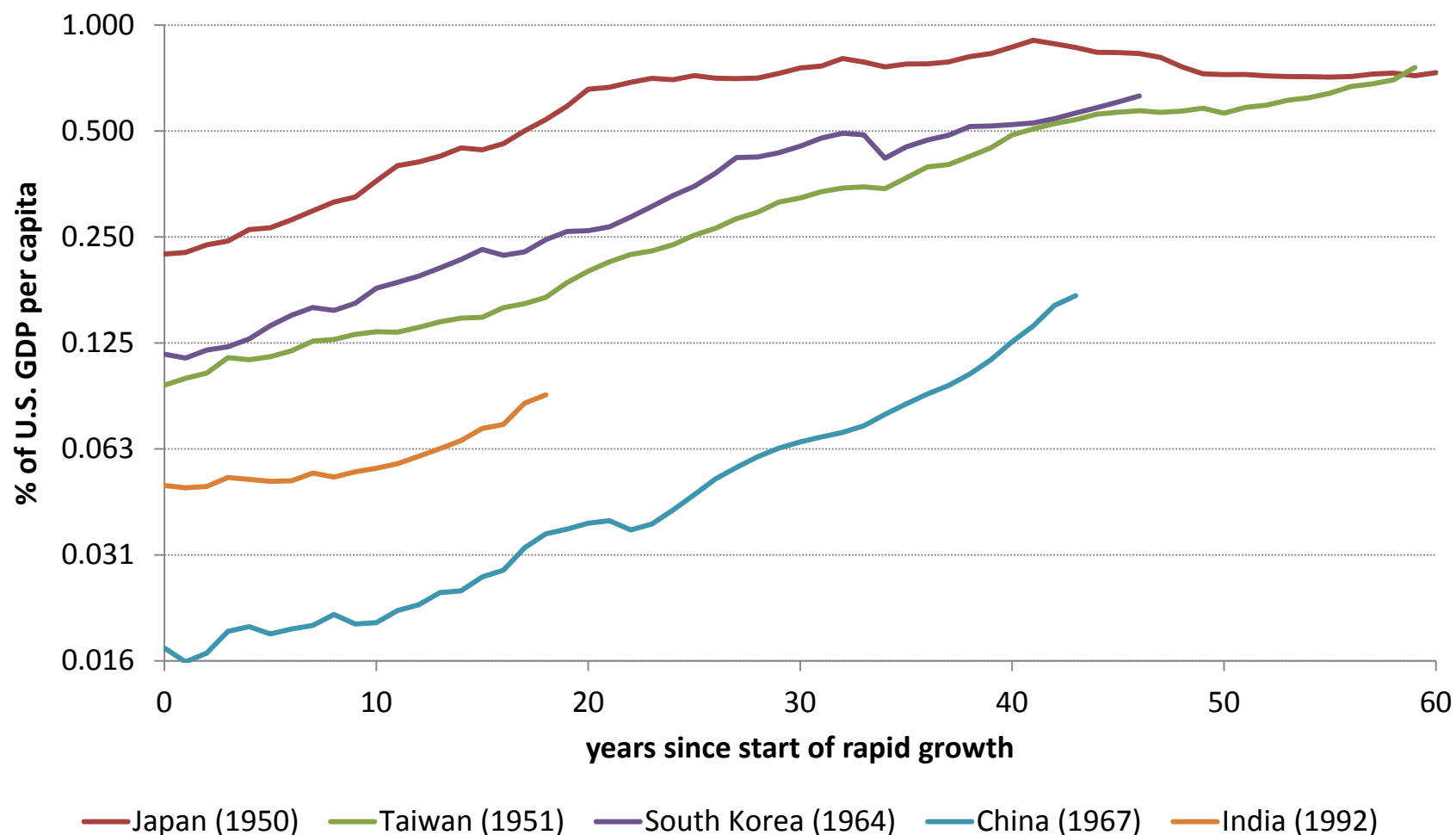


Empirical evidence – investment rates



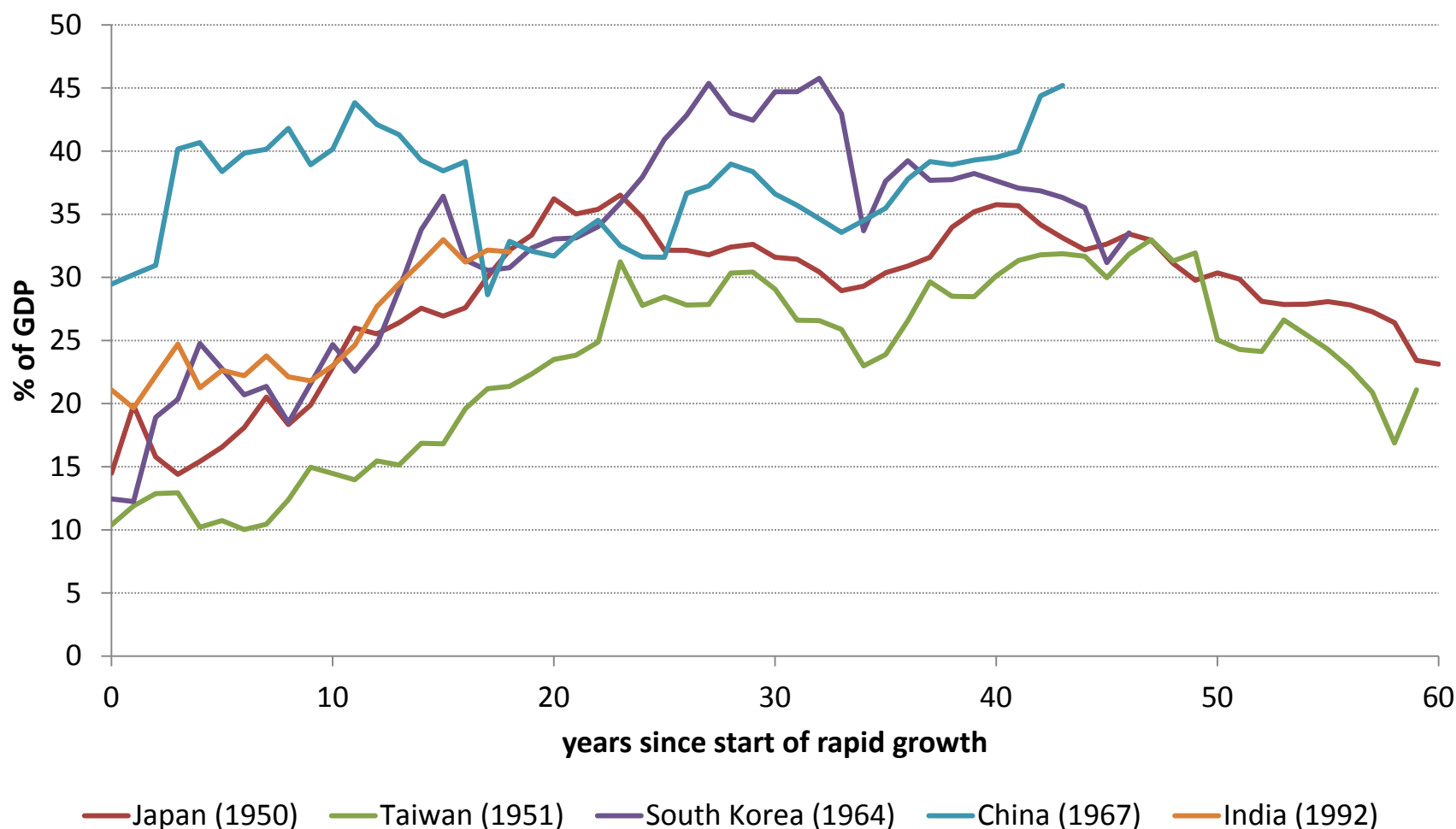
Empirical evidence – growth periods

► Year 0 = start of the high-growth period



Empirical evidence – investment rates

- ▶ Year 0 = start of the high-growth period



United States vs. China

▶ United States

- ▶ GDP per capita grows roughly 2% per year.
- ▶ This rate is about the same for more than a century.
- ▶ Interpretation: U.S. is in a “steady state”, and the steady state moves over time (this is called **balanced growth path**).
- ▶ See in the following lectures.

▶ China (and other growth miracles)

- ▶ Growth at 8-10% for more than two decades.
- ▶ China started below the steady state and is converging.
- ▶ Will growth in China slow down? Ultimately yes.
- ▶ Think about Japan after WWII – it caught up with developed world and growth slowed to roughly the same rate as the rest of developed world.

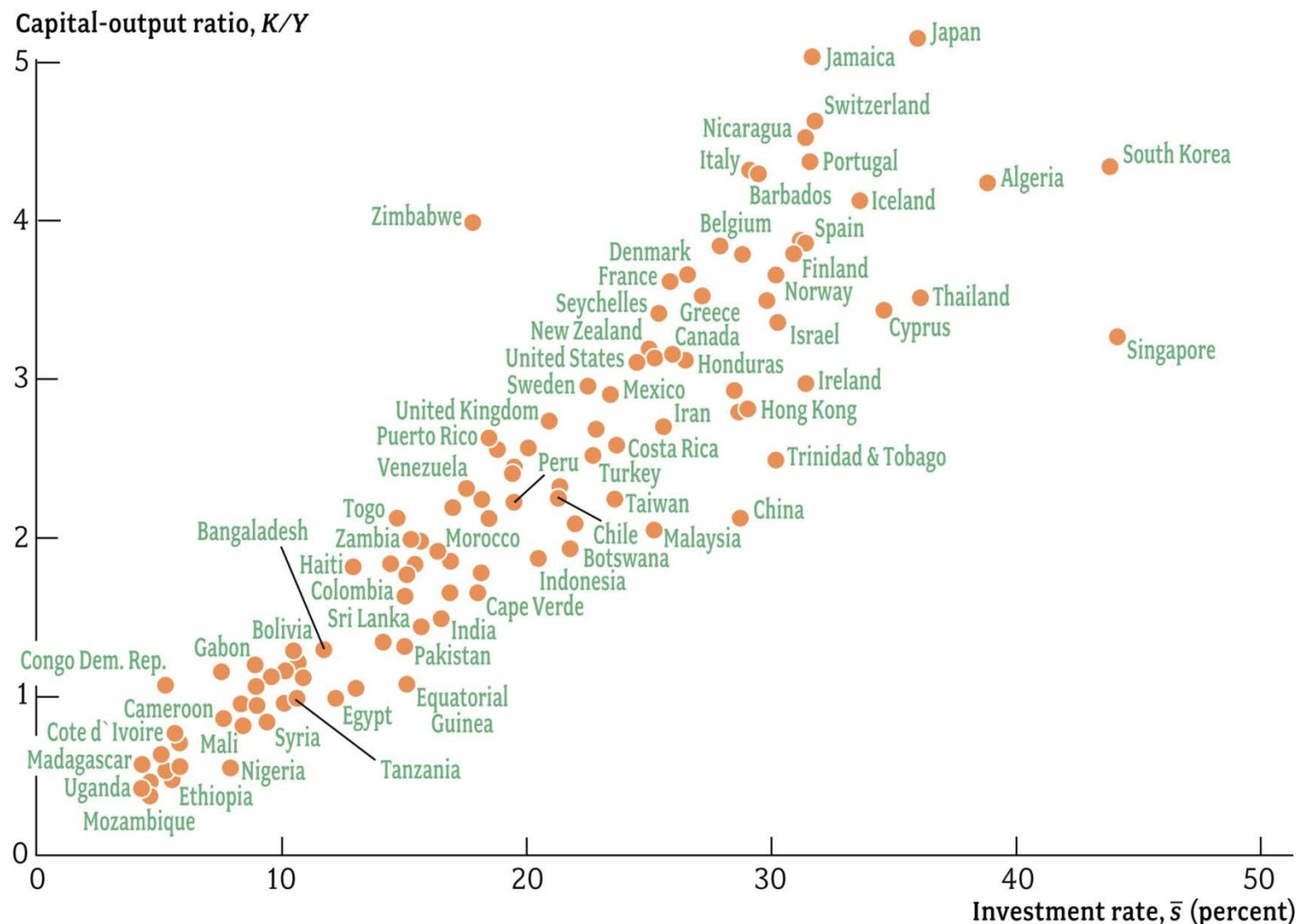
Capital-output ratio

- ▶ Recall the steady state relationship $sY^* = \delta K^*$
- ▶ Solow growth model has a prediction for the capital-output ratio

$$\frac{K^*}{Y^*} = \frac{s}{\delta}$$

- ▶ Countries with high saving rates should have more capital relative to their output.
- ▶ Why? They are willing to save more, which means that they are willing to accept a lower return on their investment.
- ▶ The last unit of capital they use will be less productive.
- ▶ Capital-output ratio will be higher.

Capital-output ratio in the data



Explaining differences in GDP per capita

- ▶ Recall steady state capital per capita: $k^* = \frac{K^*}{L} = \left(\frac{sA}{\delta}\right)^{\frac{1}{1-\alpha}}$
- ▶ Output per capita $y^* = A(k^*)^\alpha$
- ▶ Substitute in, and compare five richest and five poorest economies ($\alpha = 1/3$) (derive in class)

$$\underbrace{\frac{y_{rich}}{y_{poor}}}_{\text{Ratios: } 66} = \underbrace{\left(\frac{A_{rich}}{A_{poor}}\right)^{3/2}}_{22} \underbrace{\left(\frac{s_{rich}}{s_{poor}}\right)^{1/2}}_3$$

- ▶ What are we computing? Assume depreciation rates and α are the same across countries, but saving rates differ.
 - ▶ Differences in s given by differences in capital-output ratios (graph on previous page)

Outline

1. Empirical evidence
2. Setting up the Solow growth model
3. Saving, investment, and the real interest rate
4. Solving the Solow growth model
 1. Transition dynamics
 2. Steady state analysis, comparative statics
5. Predictions of the model and empirical evidence
6. Lessons and deficiencies

What is missing from our model?

- ▶ Prices (wage, rental rate)
 - ▶ They can be reintroduced easily.
 - ▶ The profit maximizing firm will have the same first-order conditions as in the model of production.
- ▶ Household decision
 - ▶ What determines the saving rate?
 - ▶ We would need to introduce a utility-maximizing household.
 - ▶ Consumption/saving and work decisions are fundamental in macro.
 - ▶ Household trades off consumption today for consumption in the future, depending on the real rate of interest.
 - ▶ This complicates the model pretty significantly.
- ▶ Total factor productivity still unexplained
 - ▶ Steady state still depends on A , which was our “measure of ignorance”. What happens when we increase A ?

What have we learned?

- ▶ In the long run, a country's GDP per capita depends on the fraction of output invested for the future.
 - ▶ What about fraction of people working?
 - ▶ Fraction of time spent investing in human capital (schooling)?
 - ▶ Same arguments!
- ▶ Transition dynamics
 - ▶ Poor countries should catch up fast.
 - ▶ Growth rates slow down as GDP per capita approaches that of developed countries.
- ▶ Capital accumulation alone does not lead to long-run growth.
 - ▶ Diminishing marginal returns to capital defeat long-run growth.
 - ▶ Other mechanisms must be at work.
- ▶ Capital accumulation accounts only for a small fraction of difference in GDP per capita.

Review questions

- ▶ What is the growth-generating mechanism in the Solow model?
- ▶ Does the model generate sustained growth? Why or why not?
- ▶ What is the real interest rate? How is it determined in the model?
- ▶ Why do countries in the model differ in their growth rates?
- ▶ Why do poor countries (at least the disciplined ones) grow faster than rich countries? What about the rest of the poor countries?
- ▶ What is a comparative statics exercise?
- ▶ What is the transition dynamics?
- ▶ What happens in the model when part of the capital is destroyed? When saving rate or depreciation rate increase?
- ▶ What are plausible values for the depreciation and saving rates?
- ▶ What are exogenous and endogenous variables?
- ▶ What are stock and flow variables?
- ▶ How does population growth change the picture? (problem set)